Generative Self-Supervised Learning With Spectral–Spatial Masking for Hyperspectral Target Detection

Xi Chen, Yuxiang Zhang, *Senior Member, IEEE*, Yanni Dong, *Senior Member, IEEE*, and Bo Du, *Senior Member, IEEE*

***Abstract*— Deep learning (DL) has made significant progress in hyperspectral target detection (HTD) in recent years. However, the existing DL-based HTD methods generally generate numerous labeled samples for network training, which may be impure or too similar to each other. Moreover, most methods with enormous parameters are trained and tested on the same dataset, resulting in single scenario applicability and significant computational consumption issues. To solve these issues, we propose a generative self-supervised learning (GSSL) pretraining model with spectral–spatial masking (S_2M). The lightweight vision transformer (ViT) is utilized as the backbone to learn the universal feature representation of images without labeled samples. Subsequently, the pretrained model is transferred to various HTD tasks. The transfer learning model is constructed via the lightweight ViT and a fully connected (FC) layer and fine-tuned via a weighted binary cross entropy (WBCE) loss function and a small number of selected samples. We evaluate its effectiveness on four challenging hyperspectral datasets in terms of the GSSL pretraining model, the S_2M strategy, and the WBCE loss function. Our methods achieve improvements in comparison to different pretraining models, masking strategies and loss functions. And our detection results also outperform other state-of-the-art approaches.**

***Index Terms*— Generative learning, hyperspectral target detection (HTD), self-supervised learning (SSL), vision transformer (ViT).**

I. Introduction

HYPERSPECTRAL image (HSI) usually contains hundreds of spectral bands and thus provides a rich seam of spectral information [1]. And with the high spectral resolution, HSI can help to recognize ground materials with subtle spectral differences, providing a unique advantage for target detection [2]. Hyperspectral target detection (HTD) can be regarded as a process of locating and identifying specific target pixels in HSI, aiming to isolate targets of interest from various backgrounds with few priori target spectra. The development in hyperspectral imaging prompts HTD to be widely used in various fields, such as crop quality evaluation, disease progression monitoring, geological disaster detection, and oil spill detection [3], [4], [5], [6].

Over the past few decades, various HTD methods have been proposed, which can be roughly categorized into spectral matching methods, statistical hypothesis methods, and machine learning methods. The first category mainly employs the spectral matched techniques to identify targets of interest, deriving classic constrained energy minimization (CEM) [7], orthogonal subspace projection [8], and many interesting variants, including ensemble-based CEM [9] and hierarchical-based CEM [10]. The second category makes full use of statistical assumptions about targets and backgrounds to highlight targets and suppress backgrounds, constructing generalized likelihood ratio-based detectors, such as adaptive coherent estimators (ACEs) [11], adaptive matched filters (AMFs) [12], and matched subspace detectors (MSDs) [6].

Subsequently, many machine learning techniques have been successfully explored for HTD, including kernel technique [13], [14], [15], [16], sparse representation [17], [18], [19], [20], [21], tree structure model [22], [23], and so on [24], [25]. Kernel methods map the data to high-dimensional spaces to deal with the linear inseparable problem in the original feature space, such as kernel-based CEM (KCEM) [26]. Sparse representation methods use few samples from the target and background dictionaries to linearly represent HSI and explore the reconstruction residuals to obtain the detection results. For example, Cheng and Wang [27] proposed a decomposition model with background dictionary learning for HTD (DM-BDL). Zhu et al. [28] proposed learning single spectral abundance for hyperspectral subpixel target detection (LSSA) based on the sparse constraint of the abundance matrix. One typical tree structure-based method is the target detector through tree-structured encoding (TD-TSE) [23], which constructs a binary tree to segment HSI and uses the distance on the tree between the test pixel and the given target to detect targets of interest. These traditional methods are generally simple and focus on shallow features to distinguish targets and backgrounds.

Recently, deep learning (DL) has gained prominence due to its ability of learning hierarchical, abstract, and high-level feature representations for targets and backgrounds. Various methods are proposed in the HTD field. Zhang et al. [29] proposed a deep convolutional neural network (CNN) for HTD, where CNN was trained with numerous pseudo-labeled

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samples extended by the improved autoencoder and linear prediction strategy. Zhu et al. [30] proposed a two-stream convolutional network for HTD (TSCNTD), in which the priori target samples and vast generated pseudo-labeled samples using sparse representation and linear mixing strategy were employed to train two-stream convolutional network (TSCN). Qin et al. [31] introduced a vision transformer (ViT) network to learn spectral–spatial features by the pseudo-labeled samples obtained from rough detection. Gao et al. [32] proposed a spectra aggregation and separation network (SASN) with a target band random mask to solve the lack of class-balanced training sets with pseudo labels. Li et al. [33] proposed an HTD framework based on a transformer with distributed learning (HTDFormer), which capitalizes on the parallel processing capabilities of multiple workers to efficiently handle large-scale HSIs. Xie et al. [34] proposed a background learning method based on target suppression constraint (BLTSC), in which the adversarial autoencoder (AAE) was introduced for background reconstruction to learn complex features of backgrounds. Gao et al. [35] explored an auxiliary generative adversarial network (GAN) to generate additional target and background samples, which were employed to train a detector based on the discriminator of the GAN. Shen et al. [36] proposed an interpretable representation network for HTD (HTD-IRN) using a lightweight multiscale transformer, which enables adaptive learning of background subspaces and enhances nonlinear feature representation. Besides, self-supervised learning (SSL) attracts attention in the HTD domain. Yao et al. [37] proposed a self-supervised spectral matching network to learn more discriminative features. Wang et al. [38] proposed a self-supervised spectral-level contrastive learning-based HTD (SCLHTD) method to also learn discriminative features. Zhang et al. [39] proposed a deep clustering-based self-supervised method (SSDCTD) to mine the discriminative features of targets.

The above-mentioned DL-based methods are usually trained and tested on the same image scene to ensure that the network can learn effective discriminant features for specific scenarios and tasks. And DL-based models often involve extensive samples, intricate architectures, and enormous parameters. As a result, these models require much computational time, resulting in inconveniency for most researchers and applications. Moreover, such methods lack generalization, making it challenging to transfer the learned features to other scenarios and target detector (TD) tasks. To address the above issues, Shi et al. [40] proposed a semi-supervised domain adaptive few-shot learning (SDAFL)-based detector, which trains on abundant labeled data from the hyperspectral classification dataset and transfers the trained model to the target domain. Wang et al. [41] proposed a meta-learning-based HTD using a Siamese network (MLSN). This approach leverages meta-learning techniques to learn how to discriminate between similarities and differences in spectra to obtain a pretraining model. And it utilizes the cosine similarity loss for fine-tuning the detection model, allowing for adaptation to HTD tasks. However, this method still requires a larger number of labeled samples for pretraining and unlabeled samples for fine-tuning. In addition, this method leads to a worse background suppression effect without postprocessing. Despite some progress in DL-based HTD methods, the above issues still need to be further explored to achieve better HTD performance.

Inspired by the theory of SSL [42], we propose a generative SSL (GSSL) architecture with spectral–spatial masking (S_2M) for HTD, which can be labeled as generative self-supervised target detection (GSSTD). The method uses a GSSL approach based on masked image modeling (MIM) to construct a pretraining model without labeled data. And a simple fine-tuning strategy is used to transfer the pretrained model to different target detection tasks, greatly improving time efficiency. The overall architecture of GSSTD includes two parts: generative self-supervised pretraining and transferring the pretrained model to TD tasks. First, we utilize a S_2M strategy to randomly mask the original HSI from the spectral and spatial dimension to generate a new HSI, which can help to enhance the model sensitivity to the spectral variability of HSI and complete the pretraining reconstruction task. The encoder based on ViT is utilized to map the spatially unmasked pixels into a low-dimensional feature representation. The lightweight decoder is introduced to reconstruct the original image by reverse-mapping the encoded features and masked pixels. Then, the encoder with strong feature extraction capability is transferred to downstream TD tasks, and the detector of TD tasks is designed via a fully connected (FC) layer and a sigmoid activation function, which are fine-tuned with few samples. The priori target spectra are used as the target samples, and the cluster centers obtained by K-means are used as background samples. Moreover, the weighted binary cross entropy (WBCE) loss is proposed to alleviate the sensitivity to the number of clustering centers and the tendency toward the background category during fine-tuning.

The main contributions of our research are summarized as follows.

1) To reduce the demand for labeled samples, we design a GSSL pretraining model based on S_2M strategy to learn universal feature representations without labeled data. The masking strategy can help to enhance the model sensitivity to the spectral variability of HSI and complete the pretraining reconstruction task.
2) To efficiently adapt the pretrained encoder to various TD tasks, we propose a transfer learning model via a fine-tuning strategy and a WBCE loss. The WBCE loss with few priori target spectra and clustering centers can alleviate the training sensitivity to the number of clustering centers and the training tendency toward the larger category with more training samples.

II. Related Work

A. Vision Transformer

Transformer has become an increasingly popular DL model in recent years. Transformer was initially proposed by Vaswani et al. [43] in 2017 and applied in the field of natural language processing (NLP) [44]. It employs a self-attention (SA) mechanism and parallel computing [45] to enable simultaneous processing of all elements in a sequence. This allows each position to directly focus on information from other positions, facilitating better capturing of long-distance

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dependencies in the sequence. The transformer encoder is composed of a stack of multiple identical blocks. Each block includes a multihead SA mechanism (MSA) layer and an FC feed-forward network layer. The block employs a residual connection around each of the two layers, followed by layer normalization [43]. The construction of a block is shown in Fig. 1.

SA mechanism models the correlation between each position and other positions by calculating attention weights. It first uses three different weight matrices W_Q, W_K, and W_V to linearly transform the input embedding and get the query (Q), key (K), and value (V), where each row of Q, K, and V represents a sample data. The attention score can be calculated by normalizing the inner product of Q and K, which reflects the correlation between the sample data. Then, attention weights are obtained by performing softmax normalization on the attention score, which represents the importance of different positions in the input sequence given Q. Finally, the final weighted average vector is obtained by multiplying and adding V with the attention weight, which is the feature representation corresponding to the current position. The more critical or related positions obtain higher weights. The calculation for SA can be formulated as

$$\mathrm{SA} = \mathrm{attention}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (1)$$

where d_k is the scaling factor and k is the dimension of Q and K.

Transformer utilizes the MSA mechanism to process input sequences and consider multiple attention perspectives. MSA consists of multiple blocks, and each of them is composed of various groups of Q, K, and V. These groups are connected and linearly mapped through a linear transformation to fuse into a more comprehensive feature representation. MSA can be described as

$$\mathrm{MSA} = \mathrm{MultiHead}(Q, K, V) = \mathrm{Concat}(z_1, \ldots, z_h)W^O \quad (2)$$

where $z_i = \mathrm{attention}(Q_i, K_i, V_i)$, h is the number of head, and W^O is the learnable parameter matrix, used to map the vector after multihead stitching back to the original dimension.

ViT is a strong image-oriented network based on a standard transformer encoder [45], which has been widely used in various computer vision (CV) tasks such as recognition, segmentation, and object detection [46], [47], [48]. Its core steps are listed as follows. First, the input image is divided into a set of non-overlapping patches with fixed size. Then, these patches are converted into patch embedding and fed into the MSA mechanism model. At the same time, position embedding was introduced to preserve the spatial structure of the image. This approach enables the model to simultaneously explore spatial information. Some researchers have successfully applied ViT-based approaches to HSI processing fields, such as classification [49], [50], change detection [51], unmixing [52], and detection [31], [36].

Even though achieving good detection performance, the above ViT-based detection methods are still faced with the high dependence on labeled samples and single scenario applicability issues. While we use ViT as the backbone of the GSSL framework to extract features without labeled samples and transfer the pretrained encoder based on ViT to various downstream TD tasks with a fine-tuning strategy.

B. Generative SSL

SSL learns representations or features from a large amount of unlabeled data by designing an unsupervised pretext task [42]. As a branch of SSL, GSSL [53] has demonstrated unprecedented performance in the fields of NLP [42], [54] and CV [55], [56]. It utilizes image reconstruction or generation pretext task to learn representations of data and transfer to various downstream tasks [53], [57].

Recently, a generative SSL pretraining method, MIM, had a significant impact on visual self-supervised tasks, which was first proposed by Bao et al. [58] as part of the BERT pre-training of image transformers (BEiT) framework. The MIM approach involves four key stages: masking, encoder, decoder, and loss function. First, the input image is divided into regular non-overlapping patches, which are partly and randomly masked. Then, all patch embedding is mapped to discrete tokens using an encoder. This allows the encoder to capture both global and local features of the image while maintaining an understanding of the masked areas. Finally, the decoder is used to reconstruct the original image based on discrete tokens. BEiT utilizes a discrete VAE and a standard transformer backbone. Additionally, the mean square error (MSE) loss function is used to calculate the error between the reconstructed and original image for the training constraint. This GSSL pretraining method enables the encoder to learn and generate rich semantic feature representations, which have emerged. For example, masked autoencoder (MAE) [59] introduces an asymmetric encoder and decoder, where the encoder only maps unmasked image patches. A simple framework for MIM (SimMIM) [60] replaces the decoder with a linear layer and also achieves excellent results.

The pretraining models via GSSL based on MIM have been successfully transferred to the field of hyperspectral classification [61]. To our knowledge, the GSSL pretraining model for HTD tasks is less explored. Inspired by the above, we design a GSSL pretraining model with S_2M to learn universal representation without labeled samples for downstream TD tasks. Unlike previous methods, our method implements image masking at the pixel level from the spectral and spatial dimensions.

Compared to these three self-supervised methods [37], [38], [39] for HTD, which are trained and tested on the same dataset, our model exhibits stronger generalization performance, allowing for quick adaptation to various tasks through fine-tuning. Compared to the two methods [37], [39], we do not utilize any pseudo-labeled sample selection method during the pretraining process. Furthermore, compared to [38], we employ fewer samples during fine-tuning, thus enhancing time efficiency.

III. Method

Our method can be divided into two parts: generative self-supervised pretraining and transferring the pretrained model to TD tasks. The overall architecture of GSSTD is shown in Fig. 1.

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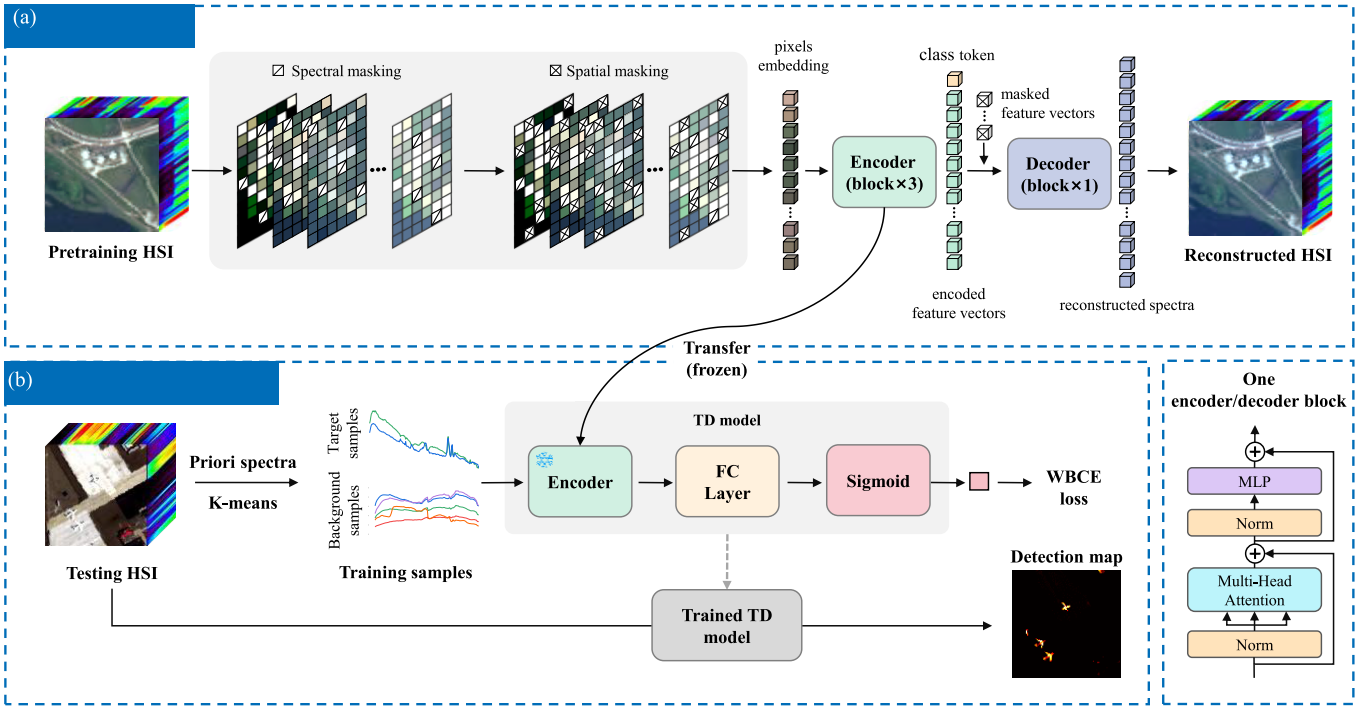


Fig. 1. Overall architecture of the proposed GSSTD method. (a) Pretraining. (b) Transferring to TD.

A. Generative Self-Supervised Pretraining

The pretraining task aims to learn the feature representation of images through the image reconstruction task. The pretraining framework is divided into four parts, namely, masking, encoder, decoder, and reconstruction error.

1) Masking: Considering the abundant spectral and spatial details, we design a S_2M strategy for pretraining model. First, we randomly mask some bands for each pixel, replacing the band values with zero. This spectral masking operation introduces significant spectral differences between different bands, thereby enhancing the model's sensitivity to spectral changes. Then, the pixels in the HSI are randomly masked from the spatial dimension, where all the entries of those spatially masked pixels' spectral vectors are set as zero. This spatial masking (SM) operation primarily serves the purpose of generating masked pixels to complete the pretraining reconstruction task. Based on the S_2M, a new HSI is constructed as input for the pretraining model. By introducing masking randomness, the pretraining model can handle different masked scenarios during each training iteration, enabling it to better adapt to variations in the input data [47].

2) Encoder: The encoder part of the pretraining framework employs ViT as the network backbone. We use a three-layer transformer with three heads to learn the feature representation. First, an FC layer is utilized to map the original pixels to pixel embeddings. The encoder takes pixel embeddings without SM as input and maps them to a low-dimensional representation, resulting in feature vectors. Then, the spatially masked pixels containing only positional information are inserted into the encoded feature vectors, enabling the input of the decoder to contain information from both masked and unmasked pixels. This feature representation utilizes the complementary information from the masked and unmasked pixels.

3) Decoder: The decoder utilizes a lightweight transformer with only one layer with four heads to reconstruct pixel information from the encoded feature vector and approach the original image. The decoder reverse-maps the encoded feature vector to restore the spatially masked pixel information. The goal of the decoder is to reconstruct the original image as accurately as possible based on the encoded features.

4) Reconstruction Error: To quantify the quality of the image reconstruction, we compare the reconstructed spectra and the original spectra of spatially masked pixels. In detail, we calculate the reconstruction error by using the MSE. The MSE loss is calculated as follows:

$$L_{\mathrm{MSE}} = \frac{1}{N}\sum_{i}^{N}\left(\hat{y}_i - y_i\right)^2 \tag{3}$$

where y_i is the original spectrum of spatially masked pixel, y'' is the reconstruction output of the decoder for spatially masked pixel, and N is the batch size. L_{MSE} is used to measure the difference between y_i and y'', which should be reduced during the training process.

B. Transfer Learning

This section demonstrates how to transfer the pretrained model to HTD task with a small number of training samples. The transfer learning model consists of four parts, namely, training sample selection, encoder, detector with fine-tuning, and WBCE loss function.

1) Training Sample Selection: In this step, we need to select a small number of samples to effectively represent the characteristics of targets and backgrounds in the testing image.

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HSI contains dominant background pixels, sparse targets, and noise. To avoid the difficult target sample selection from imbalanced data, we directly utilize the priori target spectra as target training samples, which can ensure the accuracy of target training samples. The number of priori target spectra is represented by Nt. In order to make full use of the original background information, typical background pixels obtained by K-means form background training samples. The clustering centers of different categories can simulate the variations between different background categories and guarantee the purity of background training samples. The number of clustering centers is represented by Nb. It determines the number of background training samples, which is further analyzed in the following loss function and experiment section. Such a small number of training samples can help to transfer the pretrained model to specific HTD tasks.

2) Encoder: We transfer the encoder of the pretrained model to downstream TD task. We forward the testing HSI through the encoder to exploit a low dimensional, high-level, and abstract feature, which can be used for different TD tasks.

3) Detector With Fine-Tuning: In order to detect targets of interest, we construct a detector consisting of an FC layer and a sigmoid activation function. The FC layer, with trainable parameters, can extract discriminative features and enable the model of TD tasks to predict whether each pixel of the input image belongs to the target. And by utilizing the sigmoid function, we restrict the output probabilities to a range of 0–1, thereby obtaining the final detection statistics values. During the fine-tuning process, the parameters of the encoder layer are frozen, which ensures the fine-tuning efficiency and remains the valuable feature representations learned from testing images with the pretrained encoder. Then, the parameters of the newly added FC layer are adjusted via the selected samples, aiming at better adapting the model to the specific TD task and enhancing detection performance and effectiveness.

4) WBCE Loss Function: Since the HTD task is essentially a binary classification problem, the commonly used binary cross-entropy (BCE) loss can be used to fine-tune the detector with a few target training samples and background training samples. The number of the target training samples can be set as the number of priori target spectra, which is known before detection. However, the number of background training samples needs further consideration. If the number of clustering centers is too small, the limited clustering centers will be insufficient to well represent the characteristics of various backgrounds. If the number of clustering centers is too big, there exists an imbalanced sample issue, which may lead the model to favor predicting the background category. Therefore, we take the number of target and background training samples into account and propose the WBCE loss, which is expected to alleviate the training sensitivity to the number of clustering centers and the training tendency toward the larger category with more training samples.

The WBCE loss is calculated as follows:

$$L_{\mathrm{WBCE}} = \frac{1}{N}\sum_{i=1}^{N}(1-r)\times l_i \times \log(p_i) + r \times (1-l_i)\times \log(1-p_i) \quad (4)$$

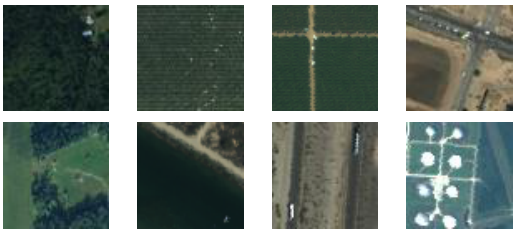


Fig. 2. Partial image scenes of the HAD100 dataset.

where $r = Nt/(Nt+Nb)$ is the ratio of target training samples to total training samples, Nt is the number of priori target spectra, and Nb is the number of clustering centers. l_i is the label of the training sample, and p_i is the output of the detector in the transfer learning stage. L_{WBCE} is used to measure the difference between l_i and p_i, which should be reduced during the fine-tuning process.

The losses generated by target training samples and background training samples are assigned different weights, and the weights are respectively set as the ratio of background training samples to total training samples and the ratio of target training samples to total training samples. The sum of two weights is one. If the number of background cluster centers is set to be larger than the number of priori target spectra, a lower weight will be assigned to the loss generated by background training samples.

Once the detector in the transfer learning stage is fine-tuned through WBCE loss, each testing image is fed into the transfer learning network for detection to obtain a detection map.

IV. Experiment

A. Experiment Details

1) Dataset Description: This work uses one hyperspectral dataset for pretraining and four hyperspectral datasets for downstream TD tasks.

HAD100 dataset [62]: The HAD100 dataset was collected by the airborne visible/infrared imaging spectrometer (AVIRIS) sensor and uniformly cropped to patches of size 64 × 64. The AVIRIS sensor collects spectral data from 224 bands ranging from 400 to 2500 nm. Fig. 2 shows eight image scenes of the HAD100 dataset. Given the abundant diversity of ground objects in this dataset, including vehicles, boats, buildings, grassland, forest, farmland, desert, river, and coast, this dataset is utilized for pretraining purposes.

San Diego dataset [10]: The San Diego dataset was collected by the AVIRIS sensor with a size of 100 × 100 pixels. After removing the water vapor absorption and noise bands, there are 189 remaining bands. It is located at San Diego airport in CA, USA. The targets of interest in this image include three aircraft, which consist of 134 target pixels. There are three priori target spectra.

Muufl dataset [63]: The Muufl dataset was collected by the compact airborne spectrographic imager (CASI)-1500 sensor with a size of 325 × 220 pixels. It is located in Long Beach, MS, USA, and is part of the Campus at the University of Southern Mississippi-Gulf Park, Hattiesburg. This dataset has 64 spectral channels in wavelengths ranging from 367.7 to

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1043.4 nm. The targets of interest in this image include several cloth boards, which consist of 269 target pixels. There are two priori target spectra.

Nuance Cri dataset [64]: The Nuance Cri dataset was collected by the Nuance Cri hyperspectral sensor with a size of 400 × 400 pixels. It is located in a lawn of Wuhan University, Wuhan, Hubei Province, China. This dataset has 46 spectral channels in wavelengths ranging from 650 to 1100 nm. The targets of interest in this image include ten rocks, which consist of 1254 target pixels. There are two priori target spectra.

Grand Isle dataset [34]: The Grand Isle dataset was collected by AVIRIS sensor with a size of 200 × 380 pixels. It is located on the Gulf Coast of Grand Isle in the part of Jefferson Parish, LA, USA. This dataset has 224 spectral channels in wavelengths ranging from 366 to 2496 nm. The targets of interest in this image are man-made objects, which consist of 279 target pixels. There are three priori target spectra.

2) Experimental Settings: In order to verify the detection performance of the proposed method GSSTD, we selected eight detection methods for comparison, including two classic detectors CEM [7] and ACE [11], three machine learning methods DM-BDL [27], TD-TSE [23] and LSSA [28], and five DL methods BLTSC [34], GAN-T [35], TSCNTD [30], HTD-IRN [36], and MLSN [41]. Through experiments and referring to the original texts, all comparison methods used the optimal parameters.

For the GSSTD method, the learning rate, the epoch, and the batch size of the pretrained network are set as 1e-4, 100, and 16, respectively. The learning rate, the epoch, and the batch size of the fine-tuning network are set as 1e-2, 200, and the number of all samples. The optimizer used is AdamW.

All the experiments are performed in the 64-bit Windows 10 operating system (OS) under the NVIDIA GeForce RTX 3060 graphics processing unit (GPU) and 16-GB random access memory.

3) Evaluation Criterion: To evaluate the detection performance, we used multiple criteria, namely, the detection map, the receiver operating characteristic (ROC) curve, the area under the ROC curve (AUC), the separability map, and the running time.

The ROC curve is plotted via segmenting the detection map with various thresholds (τ) and then obtaining different probability of detection (D) and false alarm rate (F) [21]. The horizontal coordinate represents F, and the vertical coordinate represents D. The closer the ROC curve is to the upper left corner, the better the detection performance.

A set of AUC variants are generated by the above variables D, F, and τ [65], [66]. The classical $\mathrm{AUC}_{(F,D)}$ represents the area under the ROC curve of F and D, and the higher the area value, the better the effectiveness of detector. $\mathrm{AUC}_{(\tau,F)}$ represents the area under the ROC curve of τ and F, and the lower the area value, the better the background suppression performance. $\mathrm{AUC}_{(\tau,D)}$ represents the area under the ROC curve of τ and D, and the larger the area value, the better the target detection performance. Based on the above criteria, another three AUC variants are designed to comprehensively demonstrate the performance of detectors. In detail, $\mathrm{AUC}_{\mathrm{TD}}$ represents the sum of $\mathrm{AUC}_{(F,D)}$ and $\mathrm{AUC}_{(\tau,D)}$, further indicating the target detection performance. $\mathrm{AUC}_{\mathrm{BS}}$ represents the subtract of $\mathrm{AUC}_{(F,D)}$ and $\mathrm{AUC}_{(\tau,F)}$, further indicating the background suppression performance. $\mathrm{AUC}_{\mathrm{ODP}}$ is calculated via $\mathrm{AUC}_{(\tau,D)} + (1 - \mathrm{AUC}_{(\tau,F)})$, indicating the joint target detection and background suppression performance.

The separability map is a visual tool utilized to assess separability of targets and backgrounds. It provides an intuitive evaluation of the method's performance by examining the extent of separation in the graph. Additionally, the runtime of a model serves as a crucial evaluation indicator, measuring its computational efficiency.

B. Detection Performance

This section analyzes the detection performance of the proposed GSSTD method compared with other detection methods. We evaluate the effectiveness of transferring the pretrained model to various TD tasks with four different datasets.

The detection maps of GSSTD and other comparison methods are shown in Fig. 3. CEM and DM-BDL can detect the vast majority of targets on all four datasets but cannot suppress the backgrounds well. ACE, TDTSE, BLTSC, and GAN-T exhibit strong background suppression across all four datasets, but they only detect partial targets without obvious highlighting effects. TSCNTD effectively detects most targets but misclassifies some backgrounds as targets. The reason may be that the extended target and background samples are sometimes ineffective in representativeness and discrimination. LSSA, HTD-IRN, and MLSN show prominent target detection performance on the Nuance Cri dataset but perform poorly on the other three datasets, where they fail to distinguish between targets and backgrounds. GSSTD is able to detect almost all targets, including the red panel in the lower right corner of the image scene for the Muufl dataset, the stone in the lower left corner of the Nuance Cri dataset, and the small targets in the Grand Isle dataset. These targets of discriminative difficulty are effectively separated from backgrounds with satisfactory background suppression, especially on San Diego and Grand Isle datasets. And the background suppression for Muufl and Nuance Cri datasets is expected to be further optimized when the pretrained model is fine-tuned to the best. Overall, the detection maps indicate that the proposed GSSTD can obtain better detection accuracy, more obvious target highlighting effect, and more stable background suppression.

The ROC curves for different methods on the four datasets are shown in Fig. 4. It is evident that the GSSTD method generally achieves the best ROC curves on Muufl and Grand Isle datasets, obtaining a higher probability of detection and a smaller false alarm rate. For San Diego and Nuance Cri datasets, when the false alarm rate is more than 0.001 and 0.003, the red ROC curve of GSSTD is also above other curves, indicating that GSSTD has stable detection accuracy. Furthermore, when the false alarm rate is less than 0.001, the ROC curves of three DL-based detectors (GAN-T, HTD-IRN, and MLSN) are slightly above the curve of GSSTD. The reason may be that GAN-T, HTD-IRN, and MLSN use more discriminative background samples.

The AUC values for different methods on four datasets are shown in Table I, where the highest value is displayed

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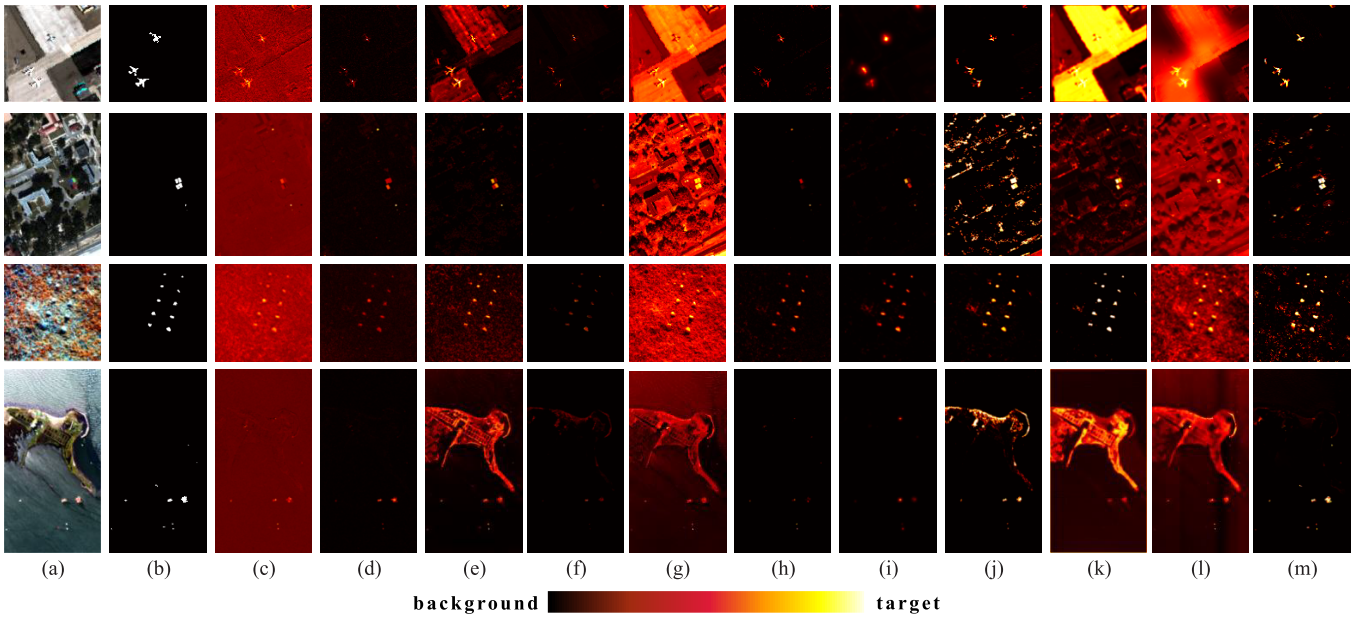


Fig. 3. Detection maps of different methods. (a) Image scenes of San Diego, Muufl, Nuance Cri, and Grand Isle datasets. (b) Grand-truth. (c) CEM. (d) ACE. (e) DM-BDL. (f) TD-TSE. (g) LSSA. (h) BLTSC. (i) GAN-T. (j) TSCNTD. (k) HTD-IRN. (l) MLSN. (m) GSSTD.

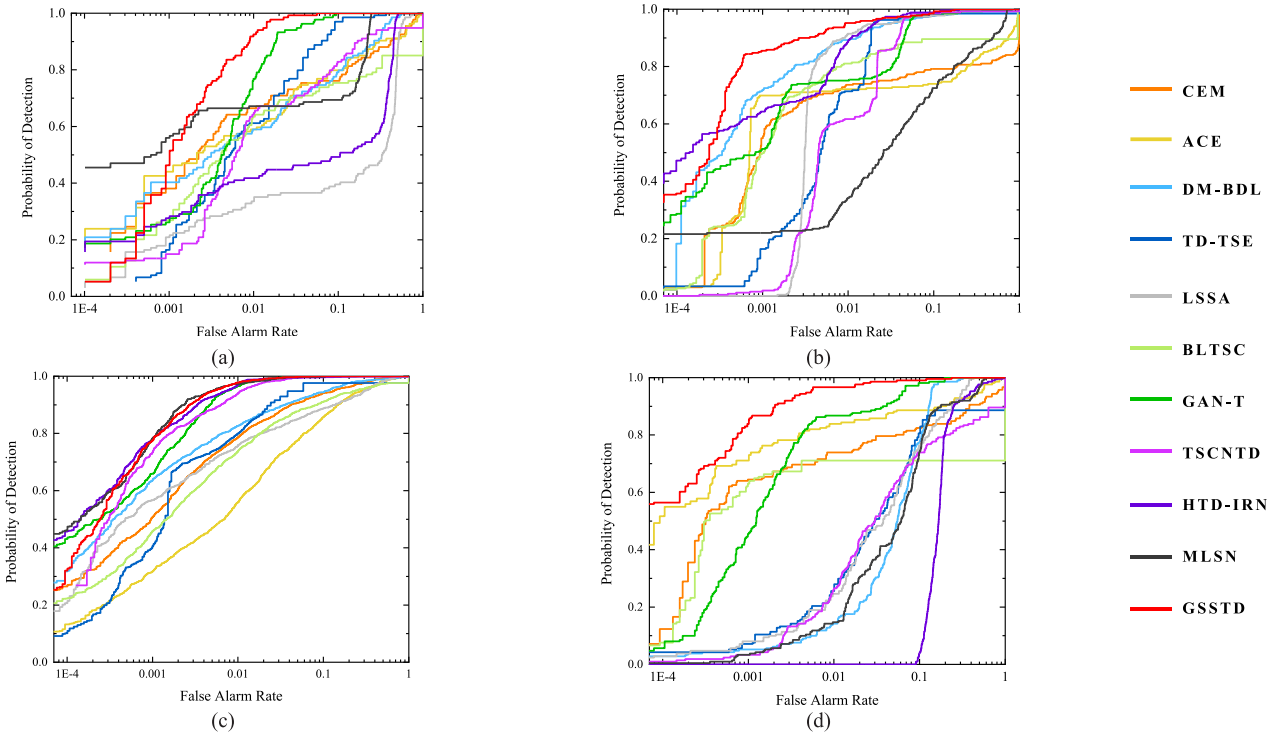


Fig. 4. ROC curves of different methods on four datasets. (a) San Diego. (b) Muufl. (c) Nuance Cri. (d) Grand Isle.

in bold. Overall, our method GSSTD performs the best on over half of the evaluation criteria. In detail, GSSTD obtains the optimal $AUC_{(F,D)}$ values on four datasets, demonstrating the effectiveness of our detection method. GSSTD obtains the optimal AUC_{ODP} values on four datasets, demonstrating the superior performance of simultaneously highlighting the target and suppressing the background. Especially, GSSTD obtains the optimal $AUC_{(\tau,D)}$ and AUC_{TD} values on three datasets, suggesting better target detection performance. GSSTD only obtains the best $AUC_{(\tau,F)}$ value on the San Diego dataset, which may indicate that the background suppression effect is slightly insufficient. However, except for the Nuance Cri dataset, GSSTD obtains the best AUC_{BS} values on all the other datasets, demonstrating a good overall background suppression performance.

The separability maps of each method for the four datasets are shown in Fig. 5, aiming to visually confirm the effectiveness of the proposed method in separating targets from

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TABLE I
AUC VALUES OF DIFFERENT METHODS FOR FOUR DATASETS

		CEM	ACE	DM-BDL	TD-TSE	LSSA	BLTSC	GAN-T	TSCNTD	HTD-IRN	MLSN	GSSTD
San Diego	$AUC_{(F,D)}$	0.9004	0.9041	0.9350	0.9787	0.7331	0.8701	0.9913	0.9307	0.8062	0.9375	**0.9967**
	$AUC_{(\tau,F)}$	0.2172	0.0122	0.0948	0.0099	0.3488	0.0024	0.0112	0.0031	0.4128	0.2912	**0.0024**
	$AUC_{(\tau,D)}$	0.4041	0.1079	0.4621	0.0992	0.5188	0.1345	0.2809	0.2345	**0.7454**	0.6436	0.4550
	AUC_{TD}	1.3045	1.0119	1.3971	1.0779	1.2520	1.0046	1.2723	1.1651	1.5516	**1.5811**	1.4516
	AUC_{BS}	0.6832	0.8919	0.8402	0.9688	0.3843	0.8677	0.9801	0.9276	0.3934	0.6463	**0.9943**
	AUC_{ODP}	1.1869	1.0957	1.3672	1.0893	1.1700	1.1321	1.2697	1.2314	1.3326	1.3524	**1.4526**
Muufl	$AUC_{(F,D)}$	0.8187	0.8520	0.9864	0.9832	0.9901	0.9015	0.9842	0.9820	0.9948	0.8800	**0.9961**
	$AUC_{(\tau,F)}$	0.1781	0.0172	0.0093	0.0006	0.2710	**0.0003**	0.0036	0.0484	0.0570	0.2050	0.0088
	$AUC_{(\tau,D)}$	0.2841	0.2669	0.3865	0.0443	0.7051	0.1112	0.2722	0.8731	0.6175	0.4160	**0.8852**
	AUC_{TD}	1.1028	1.1188	1.3728	1.0276	1.6953	1.0127	1.2563	1.8551	1.6123	1.2960	**1.8814**
	AUC_{BS}	0.6406	0.8348	0.9771	0.9827	0.7191	0.9011	0.9806	0.9337	0.9378	0.6750	**0.9873**
	AUC_{ODP}	1.1060	1.2497	1.3772	1.0438	1.4341	1.1109	1.2685	1.8247	1.5605	1.2110	**1.8764**
Nuance Cri	$AUC_{(F,D)}$	0.9767	0.9481	0.9787	0.9818	0.9577	0.9587	0.9981	0.9967	0.9982	0.9985	**0.9987**
	$AUC_{(\tau,F)}$	0.2550	0.0392	0.0605	**0.0005**	0.2835	0.0124	0.0053	0.0034	0.0080	0.1882	0.0250
	$AUC_{(\tau,D)}$	0.4842	0.2124	0.4651	0.1687	0.5970	0.3161	0.3629	0.8121	0.6136	0.6394	**0.8694**
	AUC_{TD}	1.4609	1.1605	1.4439	1.1505	1.5547	1.2748	1.3609	1.8088	1.6119	1.6380	**1.8680**
	AUC_{BS}	0.7218	0.9089	0.9182	0.9813	0.6741	0.9463	0.9928	**0.9933**	0.9902	0.8104	0.9736
	AUC_{ODP}	1.2292	1.1732	1.4046	1.1682	1.3135	1.3037	1.3576	1.8087	1.6057	1.4513	**1.8443**
Grand Isle	$AUC_{(F,D)}$	0.8974	0.9459	0.9364	0.9010	0.9240	0.8539	0.9902	0.8486	0.8127	0.9073	**0.9972**
	$AUC_{(\tau,F)}$	0.1440	0.0067	0.0575	0.0026	0.0951	**0.0001**	0.0006	0.0178	0.1151	0.1024	0.0066
	$AUC_{(\tau,D)}$	0.2849	0.2124	0.3278	0.0698	0.2797	0.0836	0.1041	0.3378	0.2013	0.3071	**0.6983**
	AUC_{TD}	1.1823	1.1583	1.2642	0.9708	1.2037	0.9375	1.0943	1.1864	1.0140	1.2144	**1.6955**
	AUC_{BS}	0.7534	0.9392	0.8789	0.8983	0.8289	0.8538	0.9896	0.8308	0.6976	0.8049	**0.9906**
	AUC_{ODP}	1.1409	1.2057	1.2703	1.0672	1.1846	1.0835	1.1036	1.3201	1.0862	1.2047	**1.6917**

backgrounds. For the last three datasets, the gap between the green box (representing the targets) and the orange box (representing the backgrounds) is consistently larger for the proposed method compared to other methods, indicating superior separability. Furthermore, the central line of green box of our method is generally positioned higher, implying that GSSTD better highlights the targets. The orange box is generally in a lower position, indicating that the proposed method can generally effectively suppress the background.

The trainable parameter amounts of the DL-based methods for four datasets are shown in Table II. This result demonstrates that MLSN and GSSTD only adjust a few parameters for different target tasks, which is significantly fewer compared to other methods in terms of training parameters. Consequently, this significantly reduces both the training time and the number of training samples. However, it is worth noting that MLSN still utilizes tens of thousands of samples for fine-tuning, while GSSTD only utilizes a dozen or so samples. As a result, even if GSSTD needs to adjust more parameters than MLSN on certain datasets, the fine-tuning time is shorter than MLSN. The training and testing time of the DL-based methods for four datasets is shown in Table III, where the optimal value is displayed in bold and the suboptimal value is underlined. MLSN and GSSTD provide the training time of fine-tuning. Due to the limited number of samples used, the proposed GSSTD method achieves more significant time savings in both training and testing, especially in terms of training time. Therefore, the GSSTD method outperforms other detection methods across all four datasets based on the evaluation criteria results. These results validate the effectiveness of transferring the GSSL pretrained model to different TD tasks.

C. Parameter Analysis and Ablation Study

1) Parameter Analysis: This section analyzes two important parameters of the proposed GSSTD method, namely, the mask ratio and the number of background samples.

First, the mask ratio for the pretraining stage was set to [10%, 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%]. The evaluation of this parameter was based on the performance of downstream target detection tasks. To mitigate the effects of random factors, five consecutive runs were performed, and the average $\mathrm{AUC}_{(F,D)}$, $\mathrm{AUC}_{(\tau,F)}$, and $\mathrm{AUC}_{(\tau,D)}$ values for four datasets are shown in Fig. 6. The San Diego, Nuance Cri, and Grand Isle datasets exhibit minor fluctuations over the whole range of mask ratio values. The Muufl dataset displays a larger fluctuation on three AUC curves, which may be attributed to the fact that serious spectral variations exist in Muufl dataset and therefore the detection performance is sensitive to the masking process. Considering the overall analysis of the four datasets, we choose a mask ratio of 50% as the uniform mask ratio.

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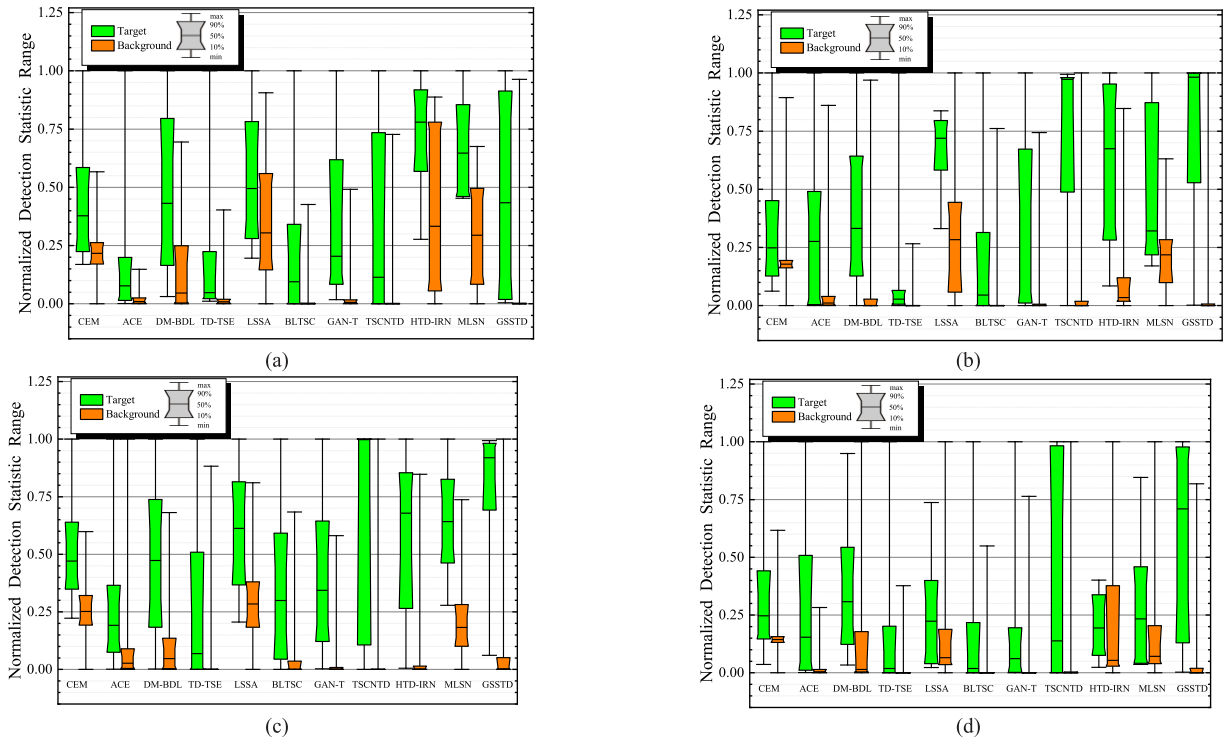


Fig. 5. Separability maps of different methods on four datasets. (a) San Diego. (b) Muufl. (c) Nuance Cri. (d) Grand Isle.

TABLE II
TRAINABLE PARAMETER AMOUNTS OF DL-BASED METHODS FOR FOUR DATASETS

	Dataset	BLTSC	GAN-T	TSCNTD	HTD-IRN	MLSN	GSSTD
Pretraining		/	/	/	/	48913	1414548
Training /Fine-tuning	San Diego	1544413	153079	45811	414709	156	193
	Muufl	1465915	88579	45811	393709	20	193
	Nuance Cri	1451497	79291	45811	390685	12	193
	Grand Isle	1594075	171139	45811	420589	210	193

TABLE III
TRAINING AND TESTING TIME (IN SECONDS) OF DL-BASED METHODS FOR FOUR DATASETS

	Dataset	BLTSC	GAN-T	TSCNTD	HTD-IRN	MLSN	GSSTD
Training /Fine-tuning	San Diego	598.2634	72.5659	264.1841	560.6545	**1.5399**	1.9719
	Muufl	1948.1477	361.3106	39.8845	3882.7332	5.1131	**1.9439**
	Nuance Cri	3504.1713	1004.9621	13.9819	7482.1441	12.0746	**1.9336**
	Grand Isle	2981.9454	553.4907	274.6592	4016.5365	7.7682	**1.9413**
Testing	San Diego	0.4869	2.1119	3.1890	0.1530	0.1782	**0.1488**
	Muufl	2.3821	0.4160	3.9561	0.7109	0.9971	**0.3969**
	Nuance Cri	6.1625	9.4719	5.6609	1.3869	2.3593	**0.8461**
	Grand Isle	11.1955	13.5059	6.2939	1.0049	1.4591	**0.5234**

Then, the number of background samples was set to [5, 10, 20, 40, 60, 80, 100, 150, 200]. The average $\mathrm{AUC}_{(F,D)}$, $\mathrm{AUC}_{(\tau,F)}$, and $\mathrm{AUC}_{(\tau,D)}$ values for four datasets are shown in Fig. 7. The $\mathrm{AUC}_{(F,D)}$ curve indicates an overall detection pattern of initial increase followed by a decrease, with the optimal performance seen at a sample size of 10. The Muufl dataset displays fluctuations for three types of AUC values over the whole range of background samples, which may also be attributed to the serious spectral variations. For the other datasets, when the number of background samples increases, $\mathrm{AUC}_{(\tau,F)}$ and $\mathrm{AUC}_{(\tau,D)}$ show a general trend of decreasing and then maintaining stability. The reason may be

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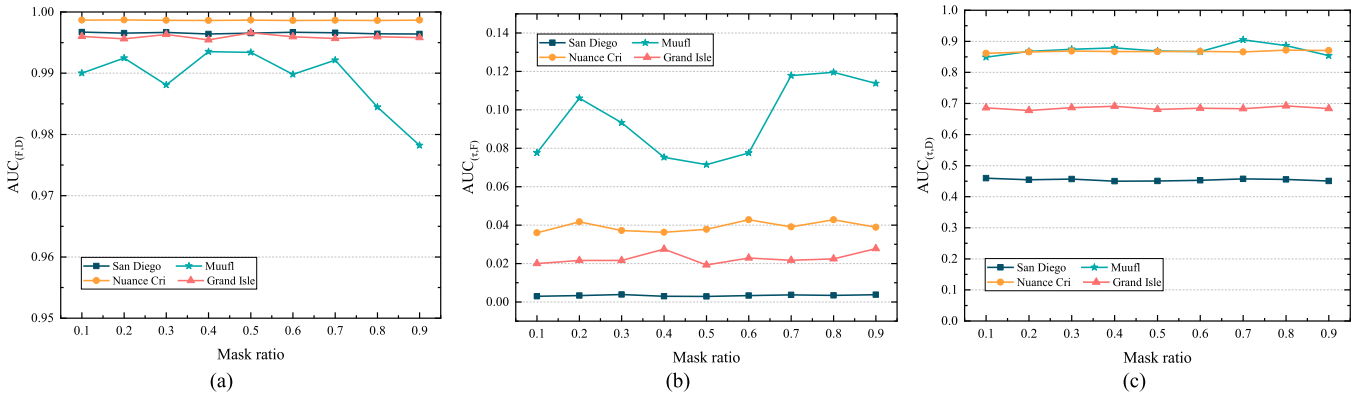


Fig. 6. Parameter analysis on four datasets for mask ratio. (a) $AUC_{(F,D)}$. (b) $AUC_{(\tau,F)}$. (c) $AUC_{(\tau,D)}$.

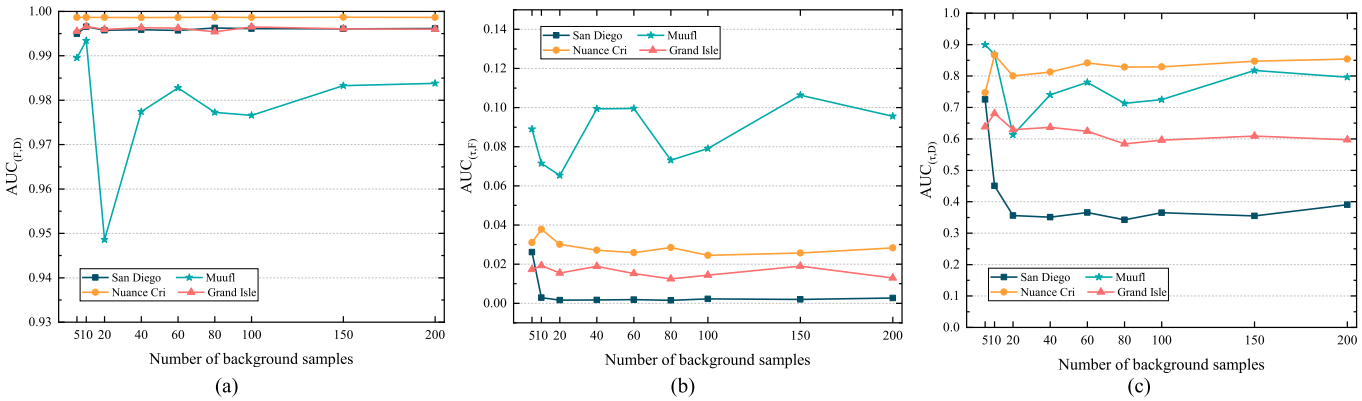


Fig. 7. Parameter analysis on four datasets for background samples. (a) $AUC_{(F,D)}$. (b) $AUC_{(\tau,F)}$. (c) $AUC_{(\tau,D)}$.

that sufficient background samples may benefit background feature learning and generate training bias, resulting in not only better background suppression but also worse target detection performance.

2) Ablation Study: This section analyzes the effectiveness of the S_2M strategy and WBCE loss in our proposed method. These models are respectively compared with SM strategy and BCE loss. In detail, we transfer the pretrained encoder with certain masking strategies for TD tasks and fine-tune the detector with certain loss. The pretext task can be formed via SM or S_2M. The target detection model can be fine-tuned via BCE loss or WBCE loss. Different choices of the masking strategy and loss function form multiple comparison experiments, which are labeled as SM-BCE, SM-WBCE, and S_2M-WBCE. To mitigate the effects of random factors, five consecutive runs were performed, and the average $AUC_{(F,D)}$, $AUC_{(\tau,F)}$, and $AUC_{(\tau,D)}$ values for four datasets are shown in Table IV.

First, a comparison between SM-BCE and SM-WBCE demonstrates significant improvements in $AUC_{(F,D)}$ and $AUC_{(\tau,D)}$ values with WBCE, indicating the advantage of WBCE in highlighting targets. A comparison between SM-WBCE and S_2M-WBCE reveals improvements in $AUC_{(F,D)}$ and $AUC_{(\tau,D)}$ values for S_2M-WBCE, suggesting that integrating spectral masking can better adapt the network to scenes with strong spectral variability. However, both SM and S2M with WBCE loss show reduced background suppression capabilities, attributed to the loss function balancing target and background samples. While there is a slight increase in $AUC_{(\tau,F)}$ values, with an average growth of only 0.0106 and 0.0043 in the four datasets, the average increments for $AUC_{(\tau,D)}$ values are 0.1389 and 0.0074, showing a stronger positive trend. Considering these results collectively, there is an overall improvement in $AUC_{(F,D)}$ results. It is important to note that despite a slight decrease

TABLE IV
AUC VALUES OF ABLATION STUDIES FOR FOUR DATASETS

Dataset		SM-BCE	SM-WBCE	S_2M-WBCE
San Diego	$AUC_{(F,D)}$	0.9964	0.9965	**0.9966**
Muufl		0.9776	0.9884	**0.9934**
Nuance Cri		0.9940	0.9984	**0.9987**
Grand Isle		0.9960	0.9965	**0.9966**
San Diego	$AUC_{(\tau,F)}$	**0.0026**	0.0028	0.0029
Muufl		**0.0712**	0.0988	0.0715
Nuance Cri		**0.0239**	0.0358	0.0378
Grand Isle		**0.0164**	0.0191	0.0192
San Diego	$AUC_{(\tau,D)}$	0.4031	0.4461	**0.4507**
Muufl		0.8090	0.8626	**0.8685**
Nuance Cri		0.4738	0.8670	**0.8672**
Grand Isle		0.6256	0.6619	**0.6808**

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in background suppression, our results still demonstrate good background suppression effect, as evident from detection maps in Fig. 3 and separability maps in Fig. 5. In conclusion, these experiments validate the effectiveness of S_2M and WBCE loss.

V. Conclusion

This article proposes a GSSTD method for HTD. By designing a generative self-supervised pretraining architecture based on S_2M, the GSSTD method can extract universal representations without labeled data for downstream tasks. And by selecting a small number of target and background samples, the transfer learning model based on the pretrained encoder and the FC layer can effectively adapt to various HTD tasks via fine-tuning strategy and WBCE loss. Extensive experiments on four challenging hyperspectral datasets verify the superiority of the GSSTD method over state-of-the-art approaches. It can better adapt to detection tasks in different scenarios and provide more accurate results while improving computational efficiency.

However, the proposed method still has a deficiency in background suppression. Even though we introduce the WBCE loss, the number of background samples still affects the detection performance, especially for the dataset with strong spectral variability, resulting in insufficient background suppression. We will explore adaptive approaches for selecting typical background samples for various datasets or even considering methods that do not generate pseudo-labeled samples, thereby further enhancing background suppression while ensuring satisfactory detection performance.

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